CAPSTONE PROJECT 03

PolicyPal: Enterprise HR and IT Policy Chatbot

Sakura Logistics, 4,000 employees in India and Japan, with HR and IT policies spread across 120 documents.

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1. The customer's problem

HR and IT help desks receive hundreds of repetitive questions a week. Employees cannot find answers in long PDFs and sometimes act on outdated information. The company wants an assistant that answers from current policies with citations, refuses when unsure, protects personal data, and hands over to a human with context.

2. Users

- Employees
- HR and IT help desk agents
- Policy owners who maintain documents
- Security and compliance team

3. User stories

- As an employee, I ask "Can I carry forward my leave?" and get a short answer with the policy section cited.
- As an employee, when the assistant cannot answer, I am handed to a human with my question summarised.
- As a policy owner, I upload an updated policy and answers change the same day.
- As security, I see that personal data is redacted and injection attempts are blocked.
- As the help desk lead, I see the most common unanswered topics each week.

4. Requirements

Must have

- Ingestion pipeline for Markdown and PDF policies with structure-aware chunking and metadata (Day 18)
- Hybrid retrieval with re-ranking and corrective RAG (Days 22, 27)
- Grounded answers with citations; refusal below a relevance threshold
- Guardrails: injection scanning on documents and inputs, PII redaction on outputs (Day 30)
- Human handoff creating a ticket with a structured payload (Day 28)
- Golden set of 30 items and a CI quality gate (Days 31, 32)
- React chat UI with citations and feedback buttons

Should have

- Re-ingestion on document change with index versioning
- Weekly report of unanswered topics and drift (Day 35)
- Japanese questions supported and evaluated

Could have (stretch)

- Role-based access to confidential policies (permission-aware retrieval, Day 18)
- Conversation memory with summaries (Day 27)
- Teams or Slack integration design

5. Data

Starter notebook writes 8 sample policy documents (leave, remote work, expenses, travel, security, IT, code of conduct, visa) and 40 example employee questions.

6. Suggested AI approach

Context engineering and chunking (Day 18), hybrid search and re-ranking (Day 22), CRAG (Day 27), guardrails (Day 30), evaluation and gates (Days 31, 32).

7. Suggested architecture

```
React UI -> FastAPI back end -> database (SQL / MongoDB)
              |-> AI service (retrieval, model, tools, guardrails)
              |-> logs, tokens and cost -> dashboard / weekly report
CI/CD: tests + security scans + quality gate -> Docker images
       -> Azure (Container Apps or App Service)
```

8. v4 baseline for this project NEW (v4)

Modernisation step: The starter notebook includes legacy_faq_answer(), a keyword if/else FAQ bot. Write characterisation tests, then replace it with the RAG service behind the same function signature, keeping the refusal behaviour.

- **AGENTS.md in the team repo:** build, test and style commands, folders not to touch, and "never put secrets in prompts" (Day 3, template in shared/templates/AGENTS_learner_template.md).
- **Security checklist** mapped to the OWASP Top 10 for Agentic Applications (shared/templates/OWASP_Agentic_Checklist.md; Days 23, 30).
- **Vulnerability report** from bandit, pip-audit, npm audit and your red-team suite (shared/templates/Vulnerability_Report_Template.md; Days 7, 20, 30).
- **Token and cost report:** tokens and estimated cost per request and per 1,000 requests, before and after one optimisation, with placeholder prices labelled (Days 12, 25, 33).

9. How to do it: day by day

Day 36 (Tue 15 Dec 2026): Kick-off and design

- Quiz on Day 35
- Teams of 3 to 4 formed; project chosen
- Problem statement, users, success metrics, assumptions log
- Architecture diagram and data model

- Repository, board and CI skeleton created, with AGENTS.md in the repo root (NEW v4)
- Pick your modernisation step and write its characterisation tests first (NEW v4)

Day 37 (Wed 16 Dec 2026): Core build

- Stand-up (15 min)
- Data loaded; back end API with main endpoints
- AI core working end to end (retrieval, recommendation, agent or pipeline)
- Golden set draft of 20 items
- Token and cost logging on every model call (NEW v4)

Day 38 (Thu 17 Dec 2026): Complete features and quality

- Stand-up
- Front end connected to the API
- Guardrails and error handling
- Evaluation run and first error analysis
- Tests in CI; quality gate added
- Modernisation step merged through a reviewed pull request (NEW v4)

Day 39 (Fri 18 Dec 2026): Harden, deploy, observe

- Stand-up
- Docker images; deploy to Azure or a documented local deployment
- Structured logs and a monitoring view
- Red-team and slice checks; fix top issues
- Security checklist (OWASP Top 10 for Agentic Applications) and vulnerability report completed (NEW v4)
- Token and cost report written (NEW v4)
- Demo rehearsal with peer feedback
- Code freeze at 17:00 IST

Day 40 (Mon 21 Dec 2026): Demo day

- Deliverables due at 09:00 IST
- Final rehearsal (morning)
- Demo: 10 minutes + 5 minutes Q&A per team
- Evaluator scoring with the rubric
- Individual reflection and peer feedback
- Closing and certificates

10. Project-specific acceptance checks

Evaluators run these during or after the demo. They count towards Core functionality and AI quality.

- Faithfulness on the golden set at least 0.9
- Unanswerable questions refused
- Injection test suite: block rate at least 90%, false alarms at most 10%
- Citations point to the right section

11. Golden set: starter items

Extend these to at least 20 items (30 for PolicyPal) with expected answers, sources and tags.

- "When does carried forward leave expire?" must include 31 March
- "Can I use any AI tool with customer data?" must say only approved tools
- "Is there a company gym?" must refuse politely

12. How you will be evaluated

Criterion	Points	What evaluators look for
Problem understanding and scoping	10	Clear problem statement, users, success metrics, assumptions log, agreed MVP scope (FDE Mindset Days 1, 14, 19).
Architecture and design	10	Sensible architecture diagram, layered code, right tool choices with reasons, data model (Days 4, 6, 18, 34).
Core functionality	20	All must-have features work end to end in the demo; project checklist items delivered.
AI and data quality	15	Golden set of at least 20 items, measured metrics (retrieval, faithfulness or recommendation quality), error analysis (Days 22, 31, 32).
Engineering quality	10	Readable code, tests (API, pipeline, prompts), Git history with pull requests, CI pipeline green (Days 15, 17, 20). NEW (v4): AGENTS.md in the repo; modernisation step done tests first.
Security, guardrails and responsible AI	10	Input and output guardrails, secrets handling, least privilege, a red-team check, fairness or language slice check (Days 7, 30). NEW (v4): OWASP Agentic checklist and vulnerability report.
Deployment and operations	10	Containerised, deployable (Azure or documented), structured logs, a dashboard or report, one drift or quality check (Days 21, 33 to 35). NEW (v4): token and cost report.
Demo and communication	10	Answer-first readout, live demo, honest limits, confident Q&A, clear handover README (FDE Mindset Days 24, 29).
Teamwork and process	5	Daily stand-ups, decision log, fair contribution, kind and specific feedback (FDE Mindset Days 14, 24, 29).

13. Common pitfalls

- No refusal path, so the bot invents policies
- Secrets or personal data in logs
- Testing only with questions copied from the documents

14. Getting started

- Open Starter_Notebooks/Capstone_03_PolicyPal_Starter.ipynb in Colab or VS Code to generate the data and see a working baseline.
- Use App_Build_Prompts/Capstone_03_PolicyPal_Prompts.md with OpenAI Codex to scaffold the project. Review every diff.

- Write your problem statement and assumptions log before writing code.

Ganbatte (頑張って): do your best